

José Giral Barajas

PHD STUDENT · MATHEMATICS RESEARCH

Imperial College London, South Kensington, London, SW7 2AZ, UK

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Education

Imperial College London

PH.D. MATHEMATICS

London, UK

2024 - present

- Advisor: Professor Paul C Bressloff

Universidad Nacional Autónoma de México (UNAM)

M.Sc. MATHEMATICS

Mexico City, Mexico

2022 - 2024

- Advisor: Dr. Sergio I. López Ortega
- Thesis: Probabilistic epidemiological model for heterogeneous populations
- Qualifying exams in partial differential equations, probability and stochastic processes, and graph theory.

Universidad Nacional Autónoma de México (UNAM)

B.Sc. APPLIED MATHEMATICS

Mexico City, Mexico

2017 - 2021

- Advisor: Dr. Sergio I. López Ortega
- Thesis: Probabilistic epidemiological model with microdynamics and reverse time (Spanish)
- First-class honours, ranked top of the and awarded the Gabino Barreda medal

Professional Experience

2021-2022 **Business Intelligence Specialist**, Totalplay Telecomunicaciones, S.A. de C.V., Mexico City, Mexico

Publications

PREPRINTS

1. **J. Giral-Barajas** and P. C. Bressloff. Instability of G/M/c queues under stochastic resetting in the interval. arXiv preprint: 2602.05009 (2026).

JOURNAL PAPERS

2. **J. Giral-Barajas** and P. C. Bressloff. Stochastic 1D search-and-capture as a G/M/c queueing model. *J. Phys. A: Math. Theor* **58** 355001 (2025).
3. **J. Giral-Barajas**, C. I. Herrera-Nolasco, M. A. Herrera-Valdez and S. I. López. A probabilistic approach for the study of epidemiological dynamics of infectious diseases: Basic model and properties. *J. Theor. Biol.* **572** 111576 (2023).

Scholarships and Awards

2024 **Roth Scholarship for Ph.D. Degree**, Department of Mathematics, Imperial College London, UK

2022 **Graduate Scholarship for M.Sc. Degree**, National Council of Humanities, Sciences and Technologies (CONAHCYT), Mexico

2022 **Gabino Barreda medal**, Universidad Nacional Autónoma de México (UNAM), Mexico

Posters

Jun 2025 Department of Mathematics PhD Symposium Poster Competition (1st prize), Imperial College London, London, UK

Nov 2023 Evento Inaugural de la Red Mexicana de Biología & Matemáticas, UNAM, Cuernavaca, Mexico

Talks and Workshops

CONTRIBUTED TALKS

- Feb 2026 Online Conference on Population Dynamics, Ecology and Evolution, *Society for Mathematical Biology, Online*
- Nov 2025 SIAM Postgraduate Conference 2025, *University of Bristol, Bristol, UK*
- Jun 2025 End-of-year Biomath Symposium, *Imperial College London, London, UK*
- Dec 2024 Non-equilibrium Systems Group seminar, *Imperial College London, London, UK*

WORKSHOPS

- Jan 2026 4th London Workshop on Active Matter, *Imperial College London, London, UK*

Teaching Experience

- Spring 2026 **Imperial College London**, Graduate Teaching Assistant, Probability and statistics
- Fall 2025 **Imperial College London**, Graduate Teaching Assistant, Probability and statistics
- Spring 2025 **Imperial College London**, Graduate Teaching Assistant, Probability and statistics
- Spring 2024 **UNAM**, Graduate Teaching Assistant, Stochastic processes
- Fall 2023 **UNAM**, Teaching Assistant, Probability I
- Spring 2022 **UNAM**, Teaching Assistant, Mathematical Modelling Workshop

Outreach

- Oct 2025 - present **Member of the SIAM Student chapter committee**, Student chapter at Imperial College London, *Imperial College London, London, UK*
- Oct 2025 **Imperial Natural Sciences blog post**, Wandering searchers and supermarket queues: mathematically exploring how cells move resources, *Imperial College London, London, UK*
- Aug 2025 **Research Fest**, Modelling Accumulation of Resources Queueing Theory and First-Passage Processes, *Imperial College London, London, UK*
- Mar 2019 **Science Fair Workshop**, Cómo calcular Pi con balines (General public), *UNAM, Mexico City, Mexico*